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ISTH SSC COMMUNICATION

Updated definition and scoring of disseminated intravascular coagulation in 2025: communication from the ISTH SSC Subcommittee on Disseminated Intravascular Coagulation

Toshiaki Iba[1] | Jerrold H. Levy[2] | Cheryl L. Maier[3] | Julie Helms[4] | Yutaka Umemura[5] | Hunter Moore[6] | Maha Othman[7,8,9] | Jecko Thachil[10] | Jean M. Connors[11] | Marcel Levi[12,13] | Ecaterina Scarlatescu[14]

[1]Department of Emergency and Disaster Medicine, Juntendo University Graduate School of Medicine, Tokyo, Japan

[2]Department of Anesthesiology, Critical Care, and Surgery, Duke University School of Medicine, Durham, North Carolina, USA

[3]Department of Pathology and Laboratory Medicine, Emory University School of Medicine, Atlanta, Georgia, USA

[4]Medical Intensive Care Unit–National Health Commission, Strasbourg University Hospital, French National Institute of Health and Medical Research, University Medical Research 1260, Regenerative Nanomedicine, Field Medical Treatment Station, Strasbourg University, Strasbourg, France

[5]Division of Trauma and Surgical Critical Care, Osaka General Medical Center, Sumiyoshi, Osaka, Japan

[6]Department of Surgery, AdventHealth Transplant Institution, Porter Hospital, Denver, Colorado, USA

[7]Department of Biomedical and Molecular Science, School of Medicine, Queen's University, Kingston, Ontario, Canada

[8]School of Baccalaureate Nursing, St Lawrence College, Kingston, Ontario, Canada

[9]Department of Clinical Pathology, Faculty of Medicine, Mansoura University, Mansoura, Egypt

[10]Department of Haematology, Manchester Royal Infirmary, Manchester, United Kingdom

[11]Hematology Division, Brigham and Women's Hospital, Harvard Medical School, Boston, Massachusetts, USA

[12]Department of Vascular Medicine, Amsterdam University Medical Center, Amsterdam, the Netherlands

[13]Department of Medicine, University College London Hospitals National Health Service Foundation Trust, Cardio-metabolic Programme-National Institute for Health and Care Research University College London Hospitals/University College London Biomedical Research Centre, London, United Kingdom

[14]Department of Anaesthesia and Intensive Care, Fundeni Clinical Institute, University of Medicine and Pharmacy "Carol Davila" Bucharest, Bucharest, Romania

Correspondence
Toshiaki Iba, Department of Emergency and Disaster Medicine, Juntendo University Graduate School of Medicine, 2-1-1 Hongo Bunkyo-ku, Tokyo 113-8421, Japan.
Email: toshiiba@juntendo.ac.jp

Abstract

Compelling evidence supports the need to update the 2001 definition and diagnosis of disseminated intravascular coagulation (DIC) to reflect our current understanding of disease pathophysiology. DIC has been long considered a critical and untreatable sequela of various underlying causes, with resolution only after treatment of the eliciting etiology. Recent views have evolved to appreciate that DIC-associated mortality may be reduced by detection and treatment at an early stage. The International Society on Thrombosis and Haemostasis Scientific and Standardization Committee on DIC proposes an updated definition of DIC: "an acquired, life-threatening intravascular disorder characterized by systemic coagulation

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activation, dysregulated fibrinolysis, and endothelial injury, resulting in microthrombosis. DIC arises from various underlying etiologies and progresses from a potentially asymptomatic early phase to an advanced phase with hemorrhage and/or organ dysfunction." In accordance with this more comprehensive definition, we propose to establish more tailored diagnostic criteria that detect early-phase DIC based on the underlying disease. We also propose a modification to overt DIC diagnostic criteria and scoring for late-phase DIC. These consensus-driven modifications reflect knowledge obtained since the original 2001 definition, which advanced clinical practice and research on DIC. This revised framework is anticipated to foster more precise and earlier diagnoses, improve patient stratification in clinical studies, and facilitate the development of targeted therapies dependent on pathophysiological context.

KEYWORDS

bleeding, capillary endothelial cell, disseminated intravascular coagulation, fibrinolysis, organ dysfunction syndrome

1 | INTRODUCTION

Clinical observations of disseminated intravascular coagulation (DIC) were initially described in the 19th century as a condition of widespread and dysregulated coagulation [1]. In the 20th century, a landmark article by Seaman [2] described DIC as an intermediary state within various diseases, highlighted by its characteristic thrombotic and hemorrhagic components. In 2001, the International Society on Thrombosis and Haemostasis (ISTH) Scientific and Standardization Committee (SSC) defined DIC as "an acquired syndrome characterized by the intravascular activation of coagulation with loss of localization arising from different causes. It can originate from and cause damage to the microvasculature, which, if sufficiently severe, can produce organ dysfunction" [3]. However, this definition fails to emphasize the dynamic progression of DIC, which ranges from subclinical to overt abnormalities in coagulation and fibrinolysis, and the importance of early detection for improving patient outcomes [4].

Many regard DIC as an untreatable decompensated coagulation disorder, and the ISTH overt DIC score remains one of the few diagnostic tools clinically available. The ISTH SSC communication also introduced diagnostic criteria for nonovert DIC; however, since these criteria included molecular biomarkers not commonly used in patient care, the nonovert DIC criteria have not been widely adopted [5]. Consequently, only the overt DIC criteria have been used in clinical practice, leading to the (mis)interpretation that DIC is synonymous with overt DIC. Moreover, a consensus on what "DIC" truly represents is needed since "DIC" does not hold a consistent meaning among researchers and physicians across the globe.

Here, the ISTH SSC on DIC proposes a revised definition based on its members' shared understanding to promote and standardize research. After the cochairs and core members of the DIC SSC had drafted this report, it was opened for input and further discussion with other SSC members and then revised. The ultimate goal of this communication is to improve patient outcomes by enabling earlier detection of DIC and allowing for more targeted management strategies.

2 | DEFINITION AND DIAGNOSTIC CRITERIA FOR DIC IN 2001 AND THE NEED FOR REVISION

In 2001, the ISTH SSC on DIC refined the definition, diagnostic criteria, and scoring systems for DIC in its communication [3]. The essential concepts of DIC include (1) DIC is an acquired syndrome involving activation of intravascular coagulation with widespread microvascular damage, (2) DIC is often triggered by critical conditions such as sepsis, trauma, and obstetric complications, resulting in both thrombosis and bleeding due to the exhaustion of coagulation factors, and (3) DIC leads to organ dysfunction when severe. The key pathophysiologic feature of DIC is that it arises from dysregulated inflammatory and coagulation responses [6]. DIC contributes to microvascular clot formation, subsequent tissue ischemia, endothelial damage, and excessive thrombin generation. In DIC, inflammation and vascular endothelial damage-induced thrombin production lead to increased microthrombi formation that impairs organ perfusion and leads to subsequent organ failure [7].

Two scoring criteria, the nonovert and DIC overt criteria, were defined in the original communication to distinguish between compensated and decompensated DIC. Although the overt DIC scoring systems use readily available laboratory tests, the lack of defined cutoff values for fibrin-related markers may lead to ambiguity in diagnosis. Another concern regarding existing criteria is the delay in diagnosis by using overt DIC criteria; specifically, it has been posited that initiating treatment based on overt DIC diagnosis is too late in the disease course, which limits management options for treating the

TABLE 1 The 2025 new International Society on Thrombosis and Haemostasis overt disseminated intravascular coagulation and sepsis-induced coagulopathy scoring systems.

	Overt DIC 2021		Overt DIC 2025	SIC
Item	Score	Range	Range	Range
Platelet count (× 10^9/L)	2	<50	<50	<100
	1	≧50, <100	≧50, <100	≧100, <150
D-dimer	3	Strong increase (or FDP)	> ×7 upper normal limit	–
	2	Moderate increase (or FDP)	> ×3 upper normal limit	–
	1	–	–	–
PT-INR	2	≧6 s	≧6 s	>1.4
	1	≧3 s, <6 s	≧3 s, <6 s	>1.2, ≦1.4 (PT-INR)
Fibrinogen (mg/dL)	1	<100	<100	–
SOFA score	2	–	–	≧2
	1	–	–	1
Total score for DIC	≧5		≧5	≧4

Total SOFA score is the sum of 4 items (respiratory SOFA, cardiovascular SOFA, hepatic SOFA, and renal SOFA). The total score of platelet count and PT-INR must exceed 2 for the diagnosis of SIC.

DIC, disseminated intravascular coagulation; FDP, fibrin/fibrinogen degradation products; INR, international normalized ratio; PT-INR, prothrombin time test-INR; PT-INT, prothrombin time prolongation/INR; SIC, sepsis-induced coagulopathy; SOFA, sequential organ failure assessment.

underlying disease and supplementing with coagulation factors and platelets [8]. The nonovert DIC scoring system requires testing for molecular markers (eg, antithrombin, protein C, and thrombin–antithrombin complexes) that are not available in many hospital laboratories. As a result, nonovert DIC scoring systems have been used primarily in research settings. Given these limitations, the ISTH SSC on DIC proposes updating the definition and diagnostic criteria for DIC in 2025.

3 | HETEROGENEITY OF CLINICAL AND PATHOPHYSIOLOGIC FEATURES OF DIC

Although they often exist concurrently, DIC presents in 2 contrasting forms: thrombotic DIC and hemorrhagic DIC, distinguished by their balance between thrombosis and bleeding [9]. From a pathophysiologic perspective, thrombotic DIC is marked by widespread microthrombi formation in microvessels. This occurs due to the excessive activation of the coagulation cascade combined with suppressed fibrinolysis, leading to microvascular thrombosis within organs and resulting in ischemia and organ dysfunction [10,11]. It should be noted that in addition to humoral factors, cellular components such as endothelial cells, platelets, and leukocytes also play a crucial role in intravascular clotting. Conversely, hemorrhagic DIC is characterized by a bleeding tendency due to the consumption of coagulation factors and platelets from widespread clotting. Hemorrhagic DIC is often a continuum of thrombotic DIC, highlighting the importance of recognizing the dynamic transition between these states. In specific conditions such as major trauma, obstetric complications, or acute promyelocytic leukemia, massive hemorrhage results not only from the depletion of clotting factors and platelets but also from extensive endogenous fibrinolysis [9,10]. In these scenarios, an initial hemorrhagic phase can evolve into a thrombotic phase. However, the role of dysregulated fibrinolysis was less emphasized in earlier definitions.

Clinically, thrombotic DIC is associated with signs of organ dysfunction caused by microvascular thrombosis, such as renal failure, liver dysfunction, and respiratory distress. In contrast, hemorrhagic DIC often presents with bleeding from multiple sites, including mucosal and puncture sites [12]. Laboratory findings further reflect these distinctions. In thrombotic DIC, elevated D-dimer is noted due to fibrin formation and breakdown, along with mild to moderate reductions in platelet count and coagulation factor levels. Hemorrhagic DIC, however, shows more pronounced thrombocytopenia, significant decreases in coagulation factors (such as fibrinogen), prolonged prothrombin time, and elevated levels of fibrin degradation products [13]. It should also be noted that other conditions can influence the laboratory results comprising the DIC score, including liver failure and thrombocytopenia from other causes. Recognizing these distinct patterns is essential for establishing appropriate diagnostic criteria.

4 | PROPOSAL OF THE REVISED DIC DEFINITION AND UPDATE FOR OVERT DIC DIAGNOSTIC CRITERIA

We propose the following updated definition for DIC: "an acquired, life-threatening intravascular disorder characterized by systemic

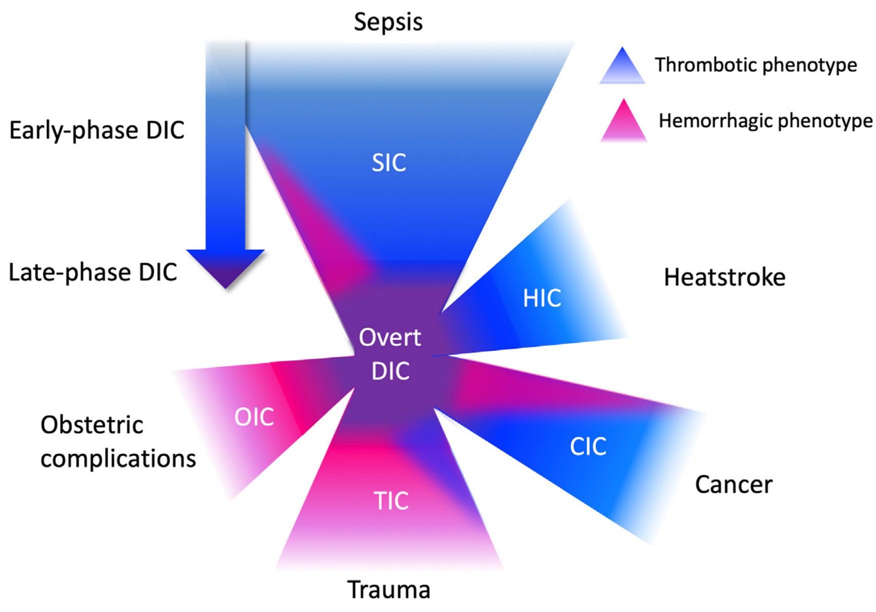

FIGURE 1 Progression from early-phase disseminated intravascular coagulation (DIC) to overt DIC. DIC arises from various backgrounds, such as sepsis, trauma, and cancer. The characteristics are different depending on the background but are largely divided into thrombotic and hemorrhagic phenotypes. Early-phase DIC finally proceeds to overt DIC along with the progression of underlying diseases. CIC, cancer-induced DIC; HIC, heatstroke-induced DIC; OIC, obstetric complication-induced DIC; SIC, sepsis-induced DIC; TIC, trauma-induced DIC.

coagulation activation, dysregulated fibrinolysis, and endothelial injury, resulting in microthrombosis. DIC arises from various underlying etiologies and progresses from a potentially asymptomatic early phase to an advanced phase with hemorrhage and/or organ dysfunction."

Although the clinical and laboratory features of late-phase DIC are common, the pathophysiology in the early phase varies significantly depending on the underlying disease [14]. Therefore, the diagnosis and management of early-phase DIC should differ and be tailored to the specific disease state [15,16]. DIC has been considered a condition where, despite different underlying causes, coagulopathy can be evaluated using one set of common diagnostic criteria. For late-phase DIC, the overt DIC diagnostic criteria can remain common. However, the current ISTH overt DIC diagnostic criteria lack clear thresholds for fibrin-related markers. Since D-dimer is more available than fibrin/fibrinogen degradation products, we propose that D-dimer be used in the overt DIC score with values of >3 times the upper normal limit be assigned 2 points and >7 times the upper normal limit be assigned 3 points, based on a previous communication of the subcommittee on DIC of the ISTH [17] (Table 1). In that report, the D-dimer cutoff for overt DIC was retrospectively evaluated with a receiver operating characteristic curve analysis in patients suspected of having DIC. Regarding D-dimer cutoffs, since D-dimer is influenced by both coagulation and fibrinolysis—and balance between these processes varies across different underlying diseases—appropriate cutoffs can differ significantly depending on the condition. Therefore, establishing universally applicable cutoffs may not be feasible. We need to know that variability and lack of standardization are drawbacks of D-dimer. Different assays and laboratory methods can give variable D-dimer results, making comparisons difficult. To improve consistency, the ISTH is trying to establish the fibrinogen equivalent unit [18]. However, so far, it is not possible to standardize the D-dimer. The other clinical laboratory tests of platelet count, fibrinogen, and prothrombin remain the same.

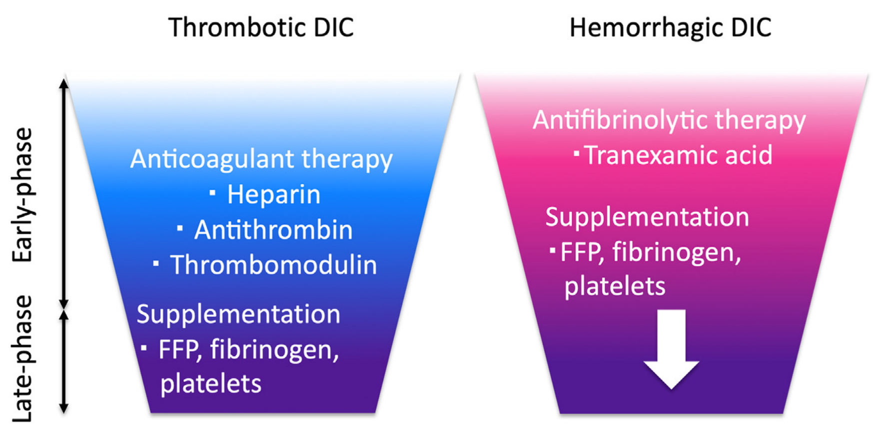

FIGURE 2 Potential treatment for disseminated intravascular coagulation (DIC). Treatment for underlying conditions is essential and mandatory. In addition, anticoagulant therapy using heparin, antithrombin, and recombinant thrombomodulin is expected for thrombotic DIC. In contrast, antifibrinolytic therapy using tranexamic acid is used for hemorrhagic DIC. In both phenotypes of DIC, supplementation of fresh frozen plasma (FFP), fibrinogen, and platelets is recommended in the late decompensated phase.

Regarding early-phase DIC, it is now understood that universal criteria are inappropriate for diagnosis, and tailored criteria should be developed for each underlying disease [19].

5 | DESIGN OF THE DIAGNOSTIC CRITERIA FOR EARLY-PHASE DIC

The critical aspect of the new DIC definitions is recognizing that DIC is a dynamic progression from early to late phase. Late-phase DIC is often complicated by clinical symptoms such as hemorrhage and organ dysfunction that are not frequently seen in early-phase DIC. Another critical point is that although hemostatic features of DIC are similar in the late stage, the initial process is different and dependent on the underlying precipitating conditions (Figure 1). Therefore, the diagnosis and management of DIC should be tailored to the specific underlying diseases.

The ISTH SSC on DIC has previously proposed diagnostic criteria for early-phase DIC arising from sepsis as sepsis-induced coagulopathy (SIC) [20,21]. The concept of SIC is that diagnostic criteria should be made with a minimum number of readily available tests [22] so that DIC can be assessed in any clinical setting. As early-phase DIC is on the continuum of DIC, diagnosis of early-phase DIC should aim to detect most cases of overt DIC at an earlier stage. Early identification enables timely intervention, which may help prevent the progression to a more severe, overt form of disease. Since its release in 2017, SIC has been widely used as the diagnostic criteria for early-phase DIC arising from sepsis [23].

However, early-phase DIC diagnostic criteria have not been developed for other nonsepsis underlying causes. Although establishing similar criteria is important, it is not an easy task. For example, in trauma, patients have a wide range of injuries with and without hemorrhagic shock and have variability of the time from injury to arrival to the hospital. Changes in coagulation in trauma patients are rapid, and they can present to emergency departments with a wide spectrum of coagulopathy, which ranges from pre-DIC with increased thrombin generation to overt DIC [24]. In massively bleeding patients, diagnosing trauma-induced coagulopathy (TIC) is not helpful in the immediate management as they require hemostatic resuscitation regardless of etiology. However, identifying patients with TIC can assist in risk stratification for progress to thrombotic-type DIC and should not be overlooked [25–27]. Comparatively, cancer-induced DIC is generally chronic, unlike the acute DIC that occurs in other critical illnesses. In cancer-associated DIC, the coagulation and fibrinolysis pathways are continuously activated but at a lower intensity. Consequently, symptoms tend to be subtle and less catastrophic compared with acute DIC. However, even in these cases, detecting and managing DIC is essential to control symptoms and maintain quality of life [28].

6 | DEFINING THE DIC-RELATED TERMINOLOGY

Consistent and precise terminology is essential for clinical practice and research on DIC. Clear definitions help ensure accurate diagnosis, effective communication among healthcare professionals, and standardized research. Precise terminology is also necessary to develop diagnostic criteria and treatment protocols and compare results across studies and clinical settings. Unfortunately, various undefined words complicate DIC research. Thus, we propose the following definitions.

6.1 | Overt DIC

This term indicates a severe, clinically evident form of hemostatic derangement characterized by widespread activation of coagulation and dysregulated fibrinolysis, manifested by organ failure and/or bleeding tendency. It leads to excessive thrombosis and an increased risk of bleeding. In overt DIC, the regulatory mechanisms are overwhelmed, resulting in the consumption of coagulation factors and platelets, bleeding, and/or organ dysfunction.

6.2 | Early-phase DIC

This term highlights the initial stage of DIC, which can progress to overt DIC if untreated. It refers to the phase where abnormal coagulation biomarkers may be detected, but major clinical findings such as bleeding, thrombotic complications, or organ dysfunction are not yet apparent, as physiologic compensatory mechanisms are able to keep up with the ongoing coagulation and fibrinolysis. Diagnosis at this stage relies on laboratory testing rather than on clinical symptoms. The significance of early-phase DIC can vary depending on the underlying cause. Nonovert DIC, subclinical DIC, and compensated DIC are synonyms for early-phase DIC, in which the patient shows subtle signs of coagulation dysfunction without the more obvious clinical symptoms seen in overt DIC.

6.3 | Pre-DIC

This term describes the earliest stage of coagulation abnormality. It refers to a situation in which the patient may have risk factors for DIC, and laboratory tests may indicate subtle abnormalities, but DIC has not yet developed. Pre-DIC was a confusing term previously used to indicate preovert DIC. Since early-phase DIC is newly defined, pre-DIC defines minor coagulopathy less than early-phase DIC.

TABLE 2 Definition of the terms related to disseminated intravascular coagulation.

DIC	
DIC is an acquired life-threatening intravascular event arising from various conditions and characterized by the systemic activation of coagulation with unbalanced fibrinolysis. It is usually accompanied by thrombocytopenia and endothelial damage. DIC can progress from a potentially nonsymptomatic early phase to an advanced phase with hemorrhage and/or organ dysfunction.	
Early-phase DIC	**Late-phase DIC**
The activated coagulation status is induced by thrombin burst and unbalanced fibrinolysis with endothelial damage. Clinical symptoms can be present but are mainly detected by laboratory tests. Anticoagulant therapy may be effective, but it has not been proven yet.	Coagulation and fibrinolysis are disrupted, and clinical symptoms such as bleeding and organ dysfunction become evident. Out-of-range anticoagulant therapy and treatments mainly involve supplementation of coagulation factors and platelets.
Synonyms: nonovert DIC subclinical DIC compensated DIC	Synonyms: overt DIC decompensated DIC
Pre-DIC	
The earlier stage of early-phase DIC. The patients have risk factors for DIC, and laboratory tests indicate very subtle abnormalities in coagulation, but DIC has not yet developed.	
Coagulopathy	
Coagulopathy is a general term for a mild to moderate impairment in 1 or more elements of coagulation. Phenotypes can be hemorrhagic, thrombotic, or both. Coagulopathy is also used in specific terms like SIC and TIC; in these situations, coagulopathy means early-phase DIC.	

DIC, disseminated intravascular coagulation; SIC, sepsis-induced coagulopathy; TIC, trauma-induced coagulopathy.

6.4 | Coagulopathy

Coagulopathy is a general term for a mild to moderate impairment in 1 or more elements of coagulation. Phenotypes can be hemorrhagic, thrombotic, or both [29]. Coagulopathy is also used in specific terms like SIC and TIC; in these situations, coagulopathy means early-phase DIC (Table 2).

7 | CONCLUSION

Currently, DIC is viewed primarily as a condition associated with poor outcomes and high mortality. For example, DIC in sepsis increases mortality by almost twice as much. However, we believe that DIC is deeply involved in the pathogenesis of various critical illnesses, and early detection and proper management can alter patient outcomes. Here, we provide an updated definition of DIC, clarify DIC-related terminology, and standardize diagnostic criteria to promote the identification of effective treatments to transform DIC into a treatable condition (Figure 2). Specifically, in this communication, we propose a revised definition and updated scoring system for DIC, emphasizing the necessity of tailoring criteria based on the underlying diseases. If we can achieve consensus and these new definitions are accepted, future clinical studies of DIC can be better harmonized with the goal of concordant diagnosis and treatment. If we can achieve consensus and these new definitions are accepted, future clinical studies of DIC can be better harmonized with the goal of concordant diagnosis and treatment.

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AUTHOR CONTRIBUTIONS

T.I., J.H.L., and E.S. wrote the draft. C.L.M., J.H., Y.U., H.M., M.O., J.T., J.M.C., and M.L. reviewed and revised the manuscript, and all of the authors reviewed and approved it.

DECLARATION OF COMPETING INTERESTS

T.I. participated on advisory boards of Japan Blood Products Organization, Asahi Kasei Pharmaceuticals, and Toray Medical. M.L. has received grants and has participated in advisory boards of Novo Nordisk, Eli Lilly, Asahi Kasei Pharmaceuticals America, and Johnson & Johnson. J.H.L. serves on the Steering or Advisory Committees for Bayer, Grifols, Octapharma, Takeda, and Werfen. E.S. received speaker's fees from Werfen, CSL Behring Germany, Prisum Healthcare Romania, and Vifor Pharma Romania. The other authors declared no conflict of interest.

ORCID

Toshiaki Iba https://orcid.org/0000-0002-0255-4088
Jerrold H. Levy https://orcid.org/0000-0003-3766-4962
Cheryl L. Maier https://orcid.org/0000-0003-4044-2674
Julie Helms https://orcid.org/0000-0003-0895-6800
Yutaka Umemura https://orcid.org/0000-0001-8357-0313
Maha Othman https://orcid.org/0000-0001-7562-203X
Jecko Thachil https://orcid.org/0000-0001-7218-0993
Jean M. Connors https://orcid.org/0000-0001-6445-582X
Marcel Levi https://orcid.org/0000-0002-2212-5299
Ecaterina Scarlatescu https://orcid.org/0000-0002-5556-3623

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